

ActInf Textbook Group ~ Cohort 6 ~ Session 5 (Chapter 2, Part 2) ~ 2/26/20

TextbookGroup/ParrPezzuloFriston2022/Cohort_6 | Cohort 6 ~ Session 5 (Chapter 2, Part 2) ~ 2/26/2024 | 57:35

all right welcome back cohort 6 Andrew
however you want to begin this
discussion on chapter two go for it sure
um probably yeah pulling up the textbook
might be a good
move
um I know that this is week two of
chapter two so I don't want to uh
reintroduce the entire chapters if you
haven't already spent a week on it but I
thought it'd be nice for anyone who
wasn't able to attend last week to just
kind of quickly skim what's going on um
so yeah this is chapter 2 the low road
to active inference in chapter one we
were introduced to the idea that there's
a high and low road to inference so um
the low road is more about the um sort
of the the the what or the the how of of
active inference and um as opposed to
the high road will where there will be
more disc discussion on the why and the
why of like why is this happening why
are agents doing this how do they
survive that's kind of the questions
getting at here this is getting much
more into sort of something like the
mechanics of the process um and it picks
up uh with the helm Holan perspective
which is a 19th century late 19th
century tradition uh that just began on
viewing uh perception as unconscious
Ence uh this idea is picked up more in
ideas such as the Basi and brain
hypothesis later on in the 20th century

and from just an act uh active inference perspective we're treating not just perception but also action planning and learning all as bayesian inference problems as well as deriving a variational approximation um for these problems to overcome intractability problems um with the helm holian tradition the kind of Distinction about this versus other historical cognitive science Traditions is that rather than viewing perception is only a bottomup process of sensory States being fed into uh being turned into internal representations of the outside or some such thing like that instead perception is viewed as an inferential process that combines topdown prior information about the most likely causes of Sensations with bottom up sensory stimuli so we actually have this as opposed to um perception being more of an outside in process it's more of a it's it's both right it's bidirectional um we're introduced to baz's rule which is just um incredibly crucial to active inference as well as just many other kinds of probability problems I strongly recommend those who are wanting to get into the maths to kind of work on um just kind of understanding baz's rule almost as like a sort of Cornerstone from which to begin um it can be viewed as a posterior uh equaling and uh a fraction that is uh the numerator is a prior and a likelihood and then your denominator is evidence and so what can happen is

that you can update any of these Quant quantities in order to solve for another quantity right so this this this can be updated over time by taking in new information that then update everything else you can update your priors your your posteriors which are your beliefs at the end like at the end of the process um and so what's nice is that we can actually map all these quantities uh onto the modeling process and active inference it's so a generative model in our case like an agent um it could be a human being it could be uh some other biological organism or otherwise the generative model is just a combination of PR uh PRIDE PRI and um likelihood mappings so priors would be something like probability of X in the textbook um in often times uh notation can also be like uh probability of O which stands for or excuse me S which stands for latent States or hidden States um likelihoods are probability of Y given X so y condition on X or in other literature can be viewed as um observations o conditioned on S hidden Latent States so we have these other quantities um involved and yeah all of those are updated using bases rule um we have some additional um information in that section box 2.1 goes over gives a brief refresher on probabilistic reasoning some Rule and product rules for anyone who wants a little bit of a who's completely you know coming into this or or maybe took a

probability course some but maybe you know it's uh the the knowledge is no longer no longer there and need a bit of a refresher um we have a nice example of an agent who confronts um you know a frog or an apple and those are kind of two example like hidden States in that particular problem right so the agent has its own beliefs about the probability it will see a frog that it will see an apple uh it's given a new observation in the moment whatever the thing is it's looking at it jumps and so that's an observation versus the S the X the hidden states which are a frog or an apple and so those get mapped together through um the the likelihood in bayes's rule as we uh as we mapped a moment ago and so the agent basically updates its beliefs about what it's seen based on the observation it just took in not sure what the thing is it could be a frog could be an apple took in a new observation it jumped uh it has its own likelihood mappings oh if it jumped then uh excuse me um yeah then it will it'll draw inference from there so we're also introduced other Concepts like surprise baseline surprise K Divergence um importance of biological plausibility um many of these theories are being being drawn from Neuroscience um and so there's there's emphasis on things like cognitive costs of computation um there's only so much one's brain can do and so whenever you consider how many computations would be

involved in you know any example beyond what we just looked at like two hidden States apples frogs um you can imagine and practice immensely more than that uh are being computed um presumably at any any given moment so um so that's where we end up looking at quantities like variational free energy uh as well as expected free energy by the end of the chapter um and this chapter I would say um this one's a really good Cornerstone for just like really getting familiar with a lot of what's going on in active inference it's it's a pretty lengthy chapter compared to the rest and um so spending some time with the different terms that are all kind of quickly being introduced here is a good idea so I guess I'll I'll kind of leave it off there and ask is anyone um you know curious about anything or more specific questions about what are going on um in in this chapter um it is a lot to say again yeah oh um so did the low road and high road come to get came did it come about like simultaneously or is it like is it just like U worked on like um like separately and then they both just came in at the same time um I do think uh Daniel would have uh a lot more knowledge on this but as far as I'm aware there's they're yeah these are coming from various traditions and are kind of a synthesis of different um yeah different Traditions

or otherwise over time and then a lot of
it is kind of uh composed primarily by
Carl Friston one of the co-authors of of
the

textbook yeah that's good answer a few
other thoughts the rhetorical device or
the metaphor was introduced by Friston
in this like beyond the desert landscape
paper it's reference somewhat theota
um any model that exists is going to
have a low road so to speak CU there's
always a how otherwise it's not being
it's just a a framework being discussed
and then any kind of a starting point
is its own kind of high road so low and
high road can apply different to to to
to probably to Frameworks other than
active

inference y um like kind of by analogy
like to almost to deny cranes and
skyhooks and like that kind of bottom up
and top down Theory and empiricism so I
think it kind of describes that
generally and then the special thing
about the low

road uh constructions and that lead to
active inference or like that are
represented by the active inference
gener model is exactly what Andrew said
which is that the Bas rule is embodied
totally by the generative
model and that describes this like broad
class of of

constructible but then compatible with
in this case free energy principle and
not necessarily something
else yeah I I guess my my thinking was
you know did it come from two minds or

was it from like one mind um using two different um perspectives to uh to think about you know um a solution to the problem I think it was many more than two overall like many people have raised many different good points about how something like has to be to persist for example

and then obviously many people have proposed different architectures and there's multiple ways to do active so it's kind of I think the you know maybe get towards higher resolution maps for the low on the high road and other things like that yep okay I thank you yeah David then Susan yeah so similar along those lines if you go back to uh the image 2.1 I think it was um between the percentage uh frogs and apples I think it was a comparison so in terms of that likelihood at what point are we talking about the same thing in terms of like if you have a belief structure of you know what percentage of apples believe they're frogs couldn't you wouldn't it be commutative in the same way where frogs would also have that same belief structure of apples if that makes sense in terms of you know occupying the same space so self- inferential are you a frog or are you an apple your prior still need to updated in terms of that correct but there would still be that percentage would that percentage not go away despite you know replacing those priors in that sense yeah that's very interesting

question so in this case this likelihood model is the a matrix in figure 4.3 because it's describing the mapping between is an apple or and what is the observation so this is kind of an exteroceptive a Matrix however the metacognitive models treat like ascending observations as observations of metacognitive observers so in that case that could refer to like interceptive data coming in observ interceptive observations or introspective observations mapping to unobserved States about what kind of thing is generating those observations like am I hungry given the stomach sensation so it could be applied internally as well to maybe some of the things you're talking about yeah I guess I was speaking more also in terms of uh what how just said in terms of uh you know of one mind or of you know more than two minds as you had mentioned okay thanks

Susan yeah so as far as um describing the low road and the high road would it be incorrect to um use a colloquial term like Tactical for low

road tactical moves does anyone think yeah go for

it yeah it's an interesting question um

I mean you know on the one hand I'm not so I'm not so sure if if that can you know be answered expressly maybe that's the kind of thing that I think someone might have to kind of do their own write up on on justifying that as kind of maybe like part of the active inference

ontology is to view it as
tactical um I mean if you were thinking
to just the context just within the
confines of this
chapter um and you're thinking about an
agent being
tactical um I would say this stuff
relating actually to maybe planning that
we see in later
sections um that might be more directly
linked just conceptually with the word
tactical in the sense of an agent is
comparing potential sequences of actions
or policies that it could take um it it
is scoring those based on the expected
free energy for each policy uh you end
up going with the one that has the
lowest expected free energy right so
that's a kind of like what should I do
here a series of paths I could take
here's the one that is least surprising
or at least is expected to have the
least amount of free energy returned
take that path right so it's almost like
strategizing now that that itself is
still kind of a loose analogy and that
you know yeah yeah I can see that yeah
so let me ask you another question that
may relate it relate it one one thought
Susan on that um Andrew that's that's a
good answer like doing the evaluation
and the the search is H is the how and
then there's other attributes of
planning that would make up like kind of
the high road of planning and I think
the stream to look at is
42 um so this is also very useful for
nested modeling and understanding the

composability of the nested modeling so here it was like analogy to strategy tactics control the center as being a higher order simpler less tactical less operational side and then their model of a robot in a warehouse has this finer scale that is like tactical that describe the posture of the robot and then a higher coarser scale that was like the location of it now this model itself could be interpreted from a low road perspective because even though this higher order is about the location of the robot it is still constructed so there's probably a lot to look at it or to think about it especially if you worked it out actually I think I think you got strategic and tactical on both ends of action and perception I mean that's what kind of came to mind when he was he was talking and and and so and I don't know if this relates directly or not but when we talk about Li high road and low road policies that do they do they have distinctly different policies no they're meeting an active inference because you only build the active inference generative model that's actually doing the inference so they just come together in that constructed model yeah I think if it's useful um I've almost wanted to bring this up from an earlier question but um figure 1.2 in the chap in the textbook which is actually in chapter one page seven um this is just to illustrate like with

active inference in the center right
like these are the high and low roads to
active inference like it's not so much
that they're entirely distinct you know
things it's more about conceptually
these are two paths that you could kind
of take to kind of move towards standing
active inference right so it's like the
low road is kind of we're starting from
base theorem we're starting from
statistics numbers calculation
valuations so on and that's what this
chapter is about meanwhile the high road
is more about the free energy principle
so now we're more expressly talking
about things like like with perception
everything we've talked about so far and
scene chapter 2 we don't really see so
much actually about like an agent trying
to survive right we don't we don't quite
see that this is just like math right
like it's just doing stuff um not so
much like trying to it's only near the
end that we get mentions of like it has
preferences that go into planning and
things like that but really this is just
kind of the mechanical aspect it's yeah
the high Rogue will be more it'll give
us much more context and I also think it
makes active inference so much more
interesting when trying to think about
how how to apply it to different
situations
so
yeah
H so I guess one of the purposes of
active influence is so that we don't
have to

manually choose the pliers right
choose prior or choose
what because in I guess in real life
with organisms you know there's uh many
fires hidden fires that you know we we
don't know that lead to you know the
perceptions and decisions that you know
they they have um you know and it'll be
too much for us
to to try and emulate or
simulate

um

I

mean do we still have to generate the pr
beforehand for an
agent so that is something about active
inference that you know there's there's
been many conversations about this and
you know it can it could totally Branch
out into much bigger philosophical
questions but active inference doesn't
necessarily um have any guarantee of
what a a prior should be it doesn't you
know it's not very good whenever it
comes to origin stories um I I suppose
is one way of stating it it's more about
this is how things happen this is why
things happen I will say um if you're
coming from more of like a computational
modeling or coding perspective uh and
you'll see this much later in the
textbook in like chapter 7 for example
um but oftentimes what's done is you'll
establish some kind of vector of priors
over initial States um so you do kind of
have to hardcode those
um the thing is you can you can have the
agent you know interact with its

environment running trials and then it can in fact from trial to trial update its own priors right so so it's kind of like once you get the the motion moving it it kind of moves on its own so to speak furthermore a lot of examples whenever it comes to modeling many people just start with uniform priors like there's no there's not something there necessarily that says you have to have you know some very um you know domain knowledge specific priors or or something like unleash the agent with uniform priors it doesn't know what's going on presumably it will you know as the low road mechanics play out it will it will readjust its own priors it kind of depends on the use case is the simplest answer yep so so some um so it can um generate it on price you know if it's given um you know like observations of like RGB you know an RGB camera and maybe yeah I don't know I guess it's just some sort of you know sensory input sure yeah it it is not widely or simply available that just because like you made a active influence agent is going to do arbitrary like infinite generalization structure learning Etc like those are very exciting directions to consider the kind of beginning State and the procedures for agents that could do things like modify the number of observations they received or understand different policy spaces so yep that's definitely to be explored and then play around with with

the different components of that
separately and then bring them together
and see how things go but if you just
look at like the PDP tutorial that is
not not going to do like open-ended
structure learning whereas maybe a
certain kind of rapper around the recent
structure learning paper would have that
kind of capacity but then not
necessarily like affect or attention or
something else that's explored in
different paper so it's always very
situational which capacities are
actually brought in Because unless
there's like some kind of open-ended or
metaprogramming element it's not going
to do something that is not described by
the generative models form is is that
like a big goal of active influence
because I mean it feels like you
without that it would just be nearly
impossible to to emulate you know the
same fires that you living organisms
would have because you we organisms
would have a lot of fire like almost you
know from our perspective almost
infinite amount of
if yes and depending on
nich it's um but definitely some people
that's similar
to what they might be interested in like
developing generalization capacities
through
metacognitive um nesting or through
observers with higher
complexity that might be something kind
of related to how Wolfram is approaching
it like with the Observer Theory and all

of that and like computational
capacity

David yeah just an observation uh can
you go back to the uh the branch map
that you had with um basian and fren was

[Music]

uh was related to what Susan was asking
it had a Bas them at the bottom and it
had um active inference at the top I
believe was a tree tree map
figure one with the high and low road
yes okay yes all right yes so sorry so
yeah Fe so I guess just like visually I
construct it this way where could you
consider this if you linked base with
free energy principle that that would
you know cover a surface so in that case
if you could do that would you have Baye
over free energy or would it be free
energy over Baye or would it be
commutative if uh you can see the
structure

there maybe write it down like write it
out as a question for the chapter and
let's return to it but okay it sounds
interesting yeah it's outside of the
bounds of what we're covering so thanks
it's just hard to process in real time
but it sounds interesting okay
yeah anyone else like to we can go
totally any

Direction

with two or look ahead or look to some
other

part one more thought on the um question
about where like the priors come from
this is coming up later in chapter nine
but chapter nine is about integrating

real data so that kind of addresses some of the questions that come up about how to integrate with real systems because like it seems like pretty clearly one kind of thing to just write out a model or like sketch it out and just propose it and then it's another thing to connect it to data and use it in like empirical settings so here's kind of that model that is constructed of like the rat and the Tas and then but that's a map of the rat's cognitive territory not claiming to be a total description of the rat's physiology just the team's decision as analyzed a certain way um so it's the Observer or the experimenter who's passing what become observations to the rat but they're actions from the scientist and then the emitted actions of the Rat are observations for the scientist so it is it is possible to understand like our role around like building the generative model in terms of this kind of laboratory setting kind of consistent with the origins and the behavioral analysis and SPM like in a laboratory making a cognitive map then that map is a map of environments and then that that was kind of explored in that in a really clear and interesting Way by by this paper with ramstead Sak divid devel and friston I'll put in the live chat but they they they make very strong claims like about that kind of relationship essentially saying if this

is how you see the situation
then Fe is the best descriptor of
it so people
could descent from their conclusions or
from their um setup of the problem or or
features but it's a very strong claim
within this
setting and then the the fallacy fallacy
like the map territory fallacy is very
ly discussed like the map of San
Francisco is not the actual land or the
territory of San Francisco and there's
like the Bor his story about the Ono one
map and how that becomes not useful
because it's kind of like wrapped over
the real world so maps are always kind
of motivated by the right um course
gradings and then um so that's the map
territory fallacy
is falsely or inappropriately
articulating the map territory
relationship but then they are resp
responding to what they perceived is a
criticism of O Fe which is um once it's
acknowledged as a cognitive modeling
technique then people say well then it
is just a map we want the real territory
description and we want the real
thing so then that's like the fallacy
fallacy is their kind of term for the
overshoot it's like okay well then it
doesn't describe anything because you
can just construct whatever map you want
it's like yes you can construct whatever
map you want of San
Francisco and then what so that's kind
of interesting angle that they brought
out even a few years ago

David yeah so in terms of that I'm just trying to think of how to how to rephrase it um terms of the mapping itself I mean obviously like through periodic time change and everything but if you created the map itself like if you I'm thinking in terms of like wave function collapse right if you're if you're observing something directly it will collapse the way it function just you know as a basic overview of it but if it's oblique to your perspective but it's still within your it's not necessarily in your priors or it might have been dissolved from your prior into something else that's kind of your own inferential map which you're not collab but you're utilizing without having to you know use that extra energy or any extra energy Beyond you know just that distilled information that you have that's kind of how I reinterpreted that that sounds very interesting like there's like a primary sensory mapping that's like dealing with a collapse over like primary observations or something but then right so instead of like a singular Viewpoint of the map and all of its other points the Viewpoint comes from you know a different perspective that you're not particularly aware of at the time of your inferential connection to the map itself so the map still exists but you're not using any energy to look at it directly so it kind of mercades itself over your own

perspective okay cool add more but it sounds interesting Cheryl

okay yes hi um I've been um as you know I'm new to this but I am very much interested in the way cognition is defined so I'm my first question is does the AC of inference then just focus on observable behaviors because so much of what the brain does is implicit and is not going to actually show up as a something that an outsider could even observe so I'll stop with that question before I go

on it's a great question so I I wouldn't say it models um only observable behaviors but rather it is made clear what is observable and what is not so you can make a model that's fully observable like a chessboard where there is no difference between the observations in the hidden State like of the piece on the board so it's like a fully observable system or you can have a system where the only observation is um a binary State and then there's like very complex internal Dynamics and another angle on that um is from this paper with bongard L so here they talk about this like poly Computing which is that the Observer is doing inference on what kind of computation a material is doing so if like there was an encrypted communication to an observer who didn't know it would look like noise that's why the

observer's um situatedness is so important because we we shouldn't just assume just because it was noise to one Observer doesn't mean that there isn't something in that information stream but it was their relationship of it that it was perceived to be doing just noise generation whereas in a different reference frame that could have other information that can't be ruled out even from from any kind of noise pattern so like I would say it's it's a big philosophical and empirical question how you model like the priors and the the hyper priors and all this of observers and that just like empirically there are techniques that basian researchers use to address some of these things like testing the the influence of different prior distributions or building the initial priors from empirical data but certainly there's a lot of expressivity and probably a lot of complexity to to what you're asking about right okay that that's helpful so um active inference does not claim to be the way that this works but a way yes that's totally true it's totally true that's why it's often helpful to um differentiate between like when are we talking about like active inference as is versus with how it can be which is improving and learning active inference or active inference versus an alternative account of the same phenomena something that like Ryan Smith

and other researchers do do and and hopefully will happen more that's one kind of comparative mode which is a totally valid and important mode because it's important um for certain questions and situations and everything like to know like does reinforcement learning fit better than active inference in this setting there other times there's kind of epistemic value even with just developing along just an active model but is always useful to have even really radically different models like is this predictor even better than just a coin flip and then that helps calibrate it to to like more reference points than just parameter sweeps within the otim model which then doesn't allow you to make points about that in relationship something that wasn't actually explored um so I just have a question about where to put questions um because you're right I think I need to write this out to get a better idea of what it is that I'm even asking you um but it has something to do with the difference between implicit and explicit so uh where would I where would I kind of think this through on right you could add a new row and like add it to this table or you could like literally just add it anywhere and okay and then do at like active inference Institute and then it it's just can be added anywhere you type it or call yeah okay great all right thank you but yeah these are interesting questions like

does the implicit and explicit map on to
observations and hidden States
correct yeah and then is there another
implicit and explicit for modeling is
that the same one described by the
internal by the hidden States and by the
um observations not
necessarily right because at one point
I'm going to ask you about neur uh
neurons and neural development and the
the way that the brain
actually um in in in embryos grows
together mation how the these things it
doesn't seem like a basian thing um and
uh because they don't all come out
different in and you know in a in a
normal curve you you know most human
beings have their brains developed along
the same lines and so um I I just would
like to know
how active
inference what what part does it play as
you go down levels of analyses or way up
in levels of analyses
past just um
agents all all great questions on the
development line like the canalization
of development like the regularity
across individuals that's definitely a
big topic like Mike Levin and other
researchers especially have looked at
that like Wasington epigenetic landscape
and understanding how like cells come to
have morphology and then I mean you
brought up other things as well but yeah
those are
like those are all great great questions
all right thank you so

much any anyone else just go for

it yes go for

it um I thought we have a section for

cohort six questions

um yeah and yeah when I was waiting for

the textbook I I do like to put down you

know my thoughts in there as well oh

awesome yeah let's look at let's look at

some right

now they do all go to the same

underlying thing but if you wanted to be

tagged with cohort 6 thank you for

suggesting yeah put it in

here and I'll make a chapter 3

one

okay

yes what happens if you just update the

prior with the posterior instead of

using K

Divergence um do you mean like what

happens if you update your belief just

like to last observation super simply

instead of calculating anything more

complex than that yeah without using the

supplies and you know I guess this to me

kind of feels like like a learning rate

kind of thing um is it is it like that

yeah like um it's also related to Colman

filter and smoothing so it's like if

we're um getting noisy thermometer

measurements I mean we could just update

our belief just to the noisiness of the

thermometer measurement and just treat

it like perfect measurement of a

underlying process but the Dynamics of

the underlying process like we might be

able to get a way better model by just

saying well it's a sinusoid hidden State

and then we're just sampling from it so
yeah I I guess it makes sense from that
perspective but from the perspective of
you know whether it's a fog or an apple
if you see an apple jump it kind of
feels like
oh like it's um
much it could be a much stronger update
that oh it is not an apple it's actually
a fog
now yeah like in this world there's a 1%
chance that apples jump so you're
totally right that if this was a apples
never jump like a heris encoded as um a
column in this Matrix yeah then it would
go 100% so it's
like in practice you can do things like
that that truncate or just like start
with no noise on a certain thing
y um Zach data yeah great question it's
um one that also came up actually
earlier so look at the cohort 5 um video
on like reinforcement learning but the
key difference is that in active
inference we're directly talking about
the joint distribution and the
likelihood of the generative
model and preferences so pragmatic value
the way that they come into with the
picture in decision making is by
checking the difference between the
preferences and the observations coming
in whereas in reinforcement learning
there's a secondary reward distribution
that's proposed on top of the
observation so like 37 degrees body
temperature is rewarding and then you
pursue rewarding

policies versus in active inference 37
is expected and preferred C and then
likely policies are
sampled that doesn't mean any given
active model is as good or better or
anything at all at all so it's totally
situational which one is more useful
which one you're more comfortable with
and so
on but that's the
difference I heard someone Susan said I
heard someone say active is based on
causality not correlation
well you can use it to study
correlations amongst variables you could
also use it like a basian causal draft
to study so-called causation amongst
variables
so then you could engage in the whole is
the causative map the territory's
causation that's a whole area like Mel
Andrews and others have explored some of
that
philosophy um and Judea Pearl kind of
and and other people delman in the
statistical causal
modeling with um Dynamic causal modeling
like if there's a causal relationship
identified in the hidden State between
two brain regions in the fmri study does
that mean they have a causal
relationship like those
are questions that there's a lot of
discussion
on
David yeah along those lines how would
that relate to like a co- causal
State co- causal State um could be

like here there's two distributions
here's temperature and humidity and they
both and then the output is the D point
so two things that jointly cause a third
thing right I'm also thinking in terms
of like a third that causes still a
singular in terms of like a xor if you
were thinking in terms of like a gated
where it's not a choice but rather it's
you know you have two
individuals making separate decisions
one person is the patient in the you
know the case we're talking about here
right terms of the patient being doing
EEG scan or something with the doctors
priors you know at you know going down
to that
0.01 right that 1% Apple frog you know
at what point is it co- causal then does
it become a xor or does it become still
in or state
or joint
co-state very interesting I think look
at this guest stream 67 if you haven't
which talks about like noisy independent
experts and about how like with sensor
arrays that having more noisy
uncorrelated which is to say noisy
components whereas like if all the
experts agreed on a test you'd actually
have low Precision relative to one with
an uncorrelated error
profile okay perfect I'll check that out
thank you
yeah someone else had their hand
raised um okay just to clarify we want
variation variational free energy to be
as small as possible approaching zero or

negative Infinity um in practice
approaching some number value that's in
arbit units um there was like a
discussion on the Discord General about
this that people could look at but like
the free energy values itself is going
to converge to
some situation specific minimal value
just like if you were doing least
squares like sum of squares minimization
for linear regression the best fit
regression would converge to like sum
sum of squares so just a a number sum
across the squared error differences
from the linear regression but then that
like value is depending on the number of
data points and the units and all this
other like random stuff not really
random stuff but it does depend on those
things so um that's why Delta free
energy and the gradient of free energy
is used and all of the physics grounding
is actually about the gradient around
the free energy and that's why it's like
incrementally optimizable because
there's a surface that gradients on free
energy can be taken on so it's like a
landscape like a free energy landscape
there's an elevation at that point
that's like the actual
score of like q and given Y and then
there's the local slopes that are used
for
optimization and then the simplest
optimization here is like is just go in
the direction that it's going to be
lower but that doesn't really matter
what the absolute value is

and then when it's when the change in free energy is zero the model is like converged to a stable Point could be local could be Global that's the whole challenge with like model yeah this is a great question I think maybe some people have put things people outside of active inference have written about like so-called conspiracy theories and Basi and belief updating um also definitely some of these topics people have looked at like people can add links but let's continue all the other questions um generating priors I mean in practice they're specified by the person who's making the generative model there should be many priors and policies yeah to make the quote Whole iguana so this is in chapter 10 but chapter 10 is a lot like chapter one so um this is like in contrast to the research program of like let's study which which is also a very useful research program but kind of two two paths diverging one of them is like studying really narrow components like let's study mechanotransduction in this brain region of groph fruit fly and then understand very narrow features very deeply um but scattered across like the animal and and non-animal species and then in contrast like this um alternative holistic approach is model the whole uana which is like how about not just

super good on the homeostasis of temperature on the Iguana then we know nothing about this other component but like think totally about it so yeah it is very challenging to make multimodal models because it's easy to make a single like an ant colony model with just foraging is easy but then if it's like foraging and Nest construction it's hard to do that David yeah so in terms of that wouldn't it wouldn't that come back then to context free grammars and being able to um compute better in that context because at what point you know when you start using frog iguana Apple I mean isn't that giving context to the measurement itself to begin with yeah very interesting question I think in the pedagogical context you could replace it with like any other it could be the the whole onion so like in that sense it doesn't like it's not a one specific just to be like clear um but like yes the reason why that makes sense or how somebody would know what to do is reflective of their ref I guess I guess it's kind of how I'm seeing it is in terms of like if you remove all the context from everything you're if you have the statistics remaining you could apply it to you know your own inference in terms of you know seeing that statistical Dynamic amongst two other things that or name something else you know that have the same exact n values that might end up you know then being if you remove the

context from it you know the statistical correlation still exists yeah there's there's a ton of things to say about that I mean there are priors I suppose that you're bringing into that as well cool let's explore it more but yes it sounds interesting all right thanks Thomas for the live streams on the noisy experts that was 67.1 I think 67.1

ntqr

Susan asks is degrees of freedom computed in act INF and not reinforcement that is not the case degrees of freedom are in place anytime statistics is being used so even if you were to use linear regression or reinforcement learning or neural network or llm

Transformer any of those statistical settings degrees of freedom are in play Anova test every single statistic setting with the meaning of what they describe the degrees of freedom of

not sure if I'm using the right term are these equations proven either mathematically or empirically or are they useful

approximations it's a complex question like in one sense all of the consistent spaces in math sub subspaces and so on like by virtue of the equal sign they're all describing

topologies all of them could be reduced to like 1 equals 1 which could come back to like the axioms of how we are going

to be defining like numbers and equality
so there's an intrinsic logic to the
equations that describes like the fact
that it really is possible as um for
example uh yakup and some other people
have done like with equation 2.5 you
really can
go and look how the
different decompositions really of free
energy and expected free energy really
are related to each other
formally so there's an internal
consistency similar to the status of all
other like mathematical statistics or
like mathematical
physics does that provide an empirical
validity is a totally interesting
question it certainly does not provide
situational validity or
adequacy but then it's like a different
interesting question does it like what
are we looking for we say empirically
and like the live stream from just a few
days ago with Ryan
Smith um on the empirical
status is super relevant as I'm sure the
insights that is going to be like later
in the week with Ryan
Smith so like and he talked a little bit
about like his philosophy background in
this stream it was really
cool because most of his work is very
clinical and like has to do with with SC
like scientifically clinical empirical
okay any last
question so this was um chapter 2 low
road the what
slow chapter 3 kind of refreshes and

picks up on the high road with uh more
about the like transcendental
imperatives side that are not
necessarily Arch architecture
dependent so I think like in that um the
robotics work of JF clader at The
Institute he's used symbolic programming
to develop kind of an alternative low
road that's not distributional belief
based but it has the same high road
imperatives so just kind of an example
of how there can be like the same
why um same Tios same um ends could be
done with different low roads kind of
interesting topics that like come out
through chapter
three then having approached actim from
the two
sides chapter 4 goes into the formal
details of the generative model that's
the central artifact that's like what is
constructed and then chapter 5 gives a
bunch of studies and reviews on one of
the systems that's been the most studied
which is like the mamalian nervous
system so kind of just showing here's
what constructing system specific models
and linking them up looks
like and and here's what it looks like
in the several decades of study that
people have done in the mammal nervous
system just as kind of like a review SL
jumping off
point
any last SL comments or
thoughts
H um I watched a video
um with do uh Dr Jeff Beck um and I'm

not sure if I'm remembering this prop
but they were saying there's some people
trying to implement the Transformer
architecture within active
influence like does that make sense at
all or maybe I'm just remembering wrong
there's projects that are doing
everything from like active rappers
around Transformers to Transformer
rappers around llms to Transformers or
neural networks replacing specific
matrices in generative models like using
a neural network for a instead of just a
matrix with the dimensions of
observations by hidden States so there's
many possible combinations I don't know
what backck discussion it was but um he
did talk with Darius and they like
talked about markall blanket and brought
up like a lot of useful questions and
ments on this does that address it cool
cool yep right thank
you
bye